

Fluid-Solid Phase Transition using Agent-based Modeling

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Abstract

This report investigates the fluid-solid phase transition using a coarse-grained, agent-based simulation. Particles interact through a Lennard-Jones potential and move under Langevin dynamics, combining deterministic interaction forces with a stochastic thermal term representing an implicit solvent. By gradually cooling the system, a clear transition from a disordered, gas-like configuration to a locally and eventually long-range ordered crystalline structure is observed, including the formation and subsequent healing of crystal defects. The physical correctness of the simulation is validated through three independent measures: the mean particle velocity, the mean squared velocity compared against the equipartition theorem, and the radial distribution function compared against the theoretical structure of a triangular lattice. All three measures support the conclusion that the observed crystallization reflects genuine physical behavior rather than numerical artifacts, demonstrating that a simplified coarse-grained model is sufficient to reproduce the key qualitative and quantitative signatures of a fluid-solid phase transition.

Chapter 1

Introduction

Phase transitions between fluid and solid states are among the most fundamental phenomena in condensed matter physics, arising from the interplay between particle interaction energy and thermal motion. While fully atomistic simulations can capture this behavior with high accuracy, they are computationally expensive and often unnecessary when the goal is to understand the qualitative mechanisms driving crystallization rather than to model a specific material exactly.

This project investigates a simplified, coarse-grained agent-based model of this transition. Instead of individual atoms, each simulated particle represents a lumped, coarse-grained entity, interacting with its neighbors through an effective Lennard-Jones potential and moving under stochastic Brownian motion, representing the influence of an implicit solvent. This approach allows the essential physics of the fluid-solid transition, competition between interaction energy and thermal energy, to be studied without the computational cost of explicitly simulating a surrounding solvent.

The particles are governed by the Langevin equation, combining deterministic interaction forces with a stochastic thermal term and a dissipative drag term. By slowly cooling the system, the transition from a disordered, gas-like state to an ordered, crystalline solid is observed directly. To ensure that this observed behavior reflects genuine physical dynamics rather than numerical artifacts, the simulation is validated using three independent measures: the mean particle velocity, the mean squared velocity in comparison to the equipartition theorem, and the radial distribution function in comparison to the theoretical structure of a triangular lattice.

This report first introduces the theoretical foundation of the model, followed by a description of its numerical implementation and validation methodology, before presenting and analyzing the resulting crystallization behavior.

Chapter 2

Theory

In this chapter we will discuss the discretely calculated new position for each particle for each timestep. Necessary for this step are position $\vec{r}$, $\vec{v}$ and $\vec{a}$, all used in the Langevin dynamics. For a System of N particles with masses M and coordinates $\mathbf{X} = \mathbf{X}(\mathbf{t})$ resulting in the Langevin equation:

$$M\ddot{\mathbf{X}} = -\nabla U(\mathbf{X}) - \gamma M\dot{\mathbf{X}} + \sqrt{2M\gamma k_B T} \mathbf{R}(t), \quad (2.1)$$

Here $\ddot{\mathbf{X}}$ and $\dot{\mathbf{X}}$ are the acceleration and velocity, $U(\mathbf{X})$ represent the particle interaction potential with ∇ as the gradient operator such that $-\nabla U(\mathbf{X})$ is the force calculated from the particle interaction potential. γ is the dampening constant, also known as the collision frequency, T is the temperature, k_B is the Boltzmann constant and $\mathbf{R}(t)$ is a delta-correlated stationary Gaussian process with zero-mean, called Gaussian white noise, satisfying

$$\langle \mathbf{R}(t) \rangle = 0 \quad (2.2)$$

$$\langle \mathbf{R}(t) \cdot \mathbf{R}(t') \rangle = \delta(t - t') \quad (2.3)$$

with δ as the Dirac delta.

For our simplified system we define the mass of each particle to be $m = 1$ and the Boltzmann constant $k_B = 1$.

In this time discretized simulation a particle is represented by its position $\vec{r}$, velocity $\vec{v}$ and acceleration $\vec{a}$. To calculate the position for timestep t_{n+1} we first calculate the velocity, based on its acceleration due to external forces and a dampening term which is proportional to the particles velocity.

$$\vec{v}(t_{n+1}) = \vec{v}(t_n) + [\vec{a}(t_n) - \gamma \vec{v}(t_n)] \Delta t + \sqrt{2\gamma T \Delta t} \vec{\eta} \quad (2.4)$$

$$\vec{r}(t_{n+1}) = \vec{r}(t_n) + \vec{v}(t_{n+1}) \Delta t \quad (2.5)$$

$$(2.6)$$

Necessary for proper Brownian motion is the correct discretized noise term, where $\vec{\eta}$ represents values from the mean-zero gaussian with a standard deviation of $\sqrt{2\gamma T \Delta t}$. T is the bath temperature of the implicit solvent and together with γ governs the size of the random kicks.

The interaction potential chosen for this model and simulation is the general Lennard-Jones potential with the form

$$V(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right] \quad (2.7)$$

with r being the distance between two particles, ϵ is the depth of the potential well and governs the interaction strength while σ is the finite distance at which the potential is zero. The r^{-12} term models short range repulsion, the r^{-6} term models the long range van der Waals attraction.

The negative derivative of the potential, gives us the force $F(r)$

$$F(r) = -\frac{dV}{dr} = \frac{24\epsilon}{r} \left[2 \left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right] \quad (2.8)$$

Now to actually calculate the between two particles we need the norm and direction

$$\vec{F}_{ij} = F(r_{ij}) \hat{r}_{ij}, \quad \hat{r}_{ij} = \frac{\vec{r}_i - \vec{r}_j}{|\vec{r}_i - \vec{r}_j|} \quad (2.9)$$

with Newton's third law giving $F_{ij} = -F_{ji}$

Another important calculation due to the periodic boundary condition is the minimum image convention. With periodic boundaries, one has to ensure that each pair interacts via its shortest separation, either directly within the boundaries or across the boundary.

$$\vec{r}_{ij} = \vec{r}_i - \vec{r}_j - L \cdot \text{round}\left(\frac{\vec{r}_i - \vec{r}_j}{L}\right) \quad (2.10)$$

where L is the periodic box size. The key here is the rounding operation is only non-zero if the distance between the points is greater than half of the periodic box itself.

$$r_c = 2.5 \cdot \sigma \quad (2.11)$$

Working with potentials and many particles can very quickly become computationally very expensive. For far reaching potentials a lot of interactions have to be calculated even if the force in the interaction is negligible. For this reason, a maximum interaction distance r_c has been established. Without it, every timestep all particles would have to interact with all others, becoming computationally increasingly expensive.

$$r_{\text{eq}} = 2^{1/6} \cdot \sigma \quad (2.12)$$

Another worth mentioning variable is the equilibrium spacing r_{eq} , which represents the distance from a particle in which $F(r) = 0$. When the coarse grains are in equilibrium and form a crystalline structure, all particles should theoretically be at this distance. In the analysis of the results, the equilibrium distance will be used to gauge if a real crystalline structure formed or not.

In the next chapter we will talk about the design of simulation and how the theory was used to produce crystalline structures and behaviors.

Chapter 3

Methodology

3.1 Simulation Design

The goal of this project was to make a simulation framework with independent interacting agents. The simulation parameters include the number of agents/interacting particles and grid size, the simulation duration and timestep size.

When choosing the timestep, we have to balance the computational intensity over the whole simulation and the correctness of the physical model. A very small time step leads to accurate physics but might take too much time to reach equilibrium in the simulation. If however the time step is chosen too large then particles might behave nonphysical and for example are able to jump over the potential barrier and occupy the same space. Forces can also diverge if they move in one timestep, from being attracted to the potential well into a diverging potential repulsion. In the first 2 versions of the simulation I had to implement a force capping mechanism to disallow nonphysical behaviour, but with a timestep of around $dt = 0.1$ the simulation behaviour was stable.

In the beginning of the simulation when all particles are spawned, a minimum distance between points makes sure that they are far enough apart to not immediately be in the potential minimum. The ratio between grid size and number of particles should also be large enough to not force them immediately into a crystalline structure but to allow them to escape the potential of their nearest neighbor.

The main simulation loop consists of calculating the forces between points that are closer than the maximum interaction range and then calculate the next position of each point based on the forces, current velocity and current positions. Furthermore values necessary for physical verification of the simulation are saved to be analyzed and plotted after the simulation.

3.2 Validation Methods

3.2.1 Mean Velocity

Under stochastic Brownian motion, the thermal noise term acting on each particle is drawn independently from a zero-mean Gaussian distribution, meaning that no direction is preferred over any other on average.

As a result, the mean velocity across all particles, $\langle \vec{v} \rangle$, is expected to converge toward zero as the number of particles N increases, since the random contributions from individual particles increasingly cancel one another out.

This provides a simple but meaningful validation check: a system with a fixed, non-vanishing net velocity would indicate a systematic bias somewhere in the implementation,

for example an asymmetry in how forces are distributed between particle pairs, or an error in how the noise term is generated. Observing $\langle \vec{v} \rangle$ remain close to zero throughout the simulation therefore confirms that the system is not accumulating any unphysical net momentum,

consistent with a closed system evolving under a stochastic thermostat with no external directional bias.

3.2.2 Mean Squared Velocity and Equipartition

A more quantitative validation is obtained from the equipartition theorem, which states that in thermal equilibrium each velocity component of a particle should on average contribute $\frac{1}{2}k_B T$ of kinetic energy.

In two dimensions, with two independent velocity components per particle, this gives the prediction $\langle v^2 \rangle = 2k_B T/m$, which in the reduced units used in this simulation ($m = 1$, $k_B = 1$) simplifies to $\langle v^2 \rangle = 2T$.

Since this value follows directly from the fluctuation-dissipation theorem underlying the Langevin equation, it provides a strong, quantitative check on the correctness of the implementation: rather than confirming that the system behaves reasonably, it tests whether the noise strength, drag coefficient, integration scheme and collision calculations are correctly balanced to reproduce the prescribed temperature T .

Measuring $\langle v^2 \rangle$ over the course of a simulation and comparing its long time average against the theoretical value of $2T$ therefore serves as a direct test of the thermostat itself, independent of any structural or qualitative assessment of the resulting particle configuration.

3.2.3 Radial Distribution Function

The radial distribution function $g(r)$ provides a structural, rather than purely thermodynamic, validation of the simulation, capturing how particle density varies as a function of distance from any given particle relative to what would be expected in a uniform, uncorrelated gas of the same density.

A value of $g(r) = 1$ indicates no structural preference at that distance, while pronounced peaks indicate distances at which particles are systematically more likely to be found, corresponding to preferred neighbor spacings arising from the Lennard-Jones interaction.

For a crystalline configuration, these peaks are expected to align closely with the theoretical neighbor-shell distances of a triangular lattice, $r = a\sqrt{i^2 + ij + j^2}$ for integers i, j , where $a = \sigma \cdot 2^{1/6}$ is the equilibrium spacing predicted directly from the force law. Sharp, well-separated peaks that match this sequence therefore serve as direct,

independent confirmation that the system has organized into a locally hexagonal-packed solid, while broadened or absent peaks are characteristic of a liquid or gas phase lacking long-range order.

Because $g(r)$ is derived purely from particle positions rather than from velocities or forces, it validates the simulation from an entirely different perspective than the kinetic checks above, confirming that the correct microscopic interaction law is not only being applied correctly at each timestep, but is also producing the macroscopic structural behavior it is expected to produce.

Chapter 4

Results

The final step in achieving proper results and investigating the properties of the simulated system is to include a gradual cooling of the solvent and therefore slowly decreasing the strength of the Brownian motion.

In figure 4.1 we investigate the behavior of interacting particles, the strength of their coupling in comparison with their kinetic energy.

The four figures below have been generated with 360 particles, a grid size of 80σ , time step $dt = 0.05$. The general temperature of the system began at $T = 1.3$ and cooled down to $T_{min} = 0.05$ with a cooling rate of 0.0015 per time step.

The drag coefficient was set at $\gamma = 1.3$, well depth at $\epsilon = 2.4$ and particle size $\sigma = 3.0$. The simulation starts and very quickly a fully chaotic system is present where the initial state is not visible anymore as seen in figure 4.1a.

We observe no coupling, particles have enough kinetic energy to move between neighboring potential wells. The force calculation seems physical and does not allow for particles to collapse onto each other, the repulsive forces keeping them apart correctly.

In figure 4.1b we see that weak force coupling has set in. With a temperature of $T = 0.811$ particles are still able to switch places, move between different potentials but no free moving particles are visible. Local weak coupling has set in, but the mean kinetic energy is still high enough for no long-range order emerges.

Locally a semi hexagonal order has emerged but crystal defects are apparent. Vacancies, meaning empty local spaces, exist inside the solidifying structure. With a temperature $T = 0.551$ and $t = 9714$ in figure 4.1c a long range coupling is strongly visible. The vacancies have healed and filled, prior fault lines have now reduced to weakly non parallel long range fault lines.

A solidifying crystalline structure is clearly visible. All particles inside the emerged clusters have a hexagonal structure although local irregularities are still present.

Finally we have figure 4.1d in which all irregularities within clusters disappeared, long range coupling has fully set in with $T = T_{min} = 0.05$ and $t = 16992$.

The parameter choice for γ, ϵ and σ for the presented figure was chosen because it produced clearly visible phase transitions.

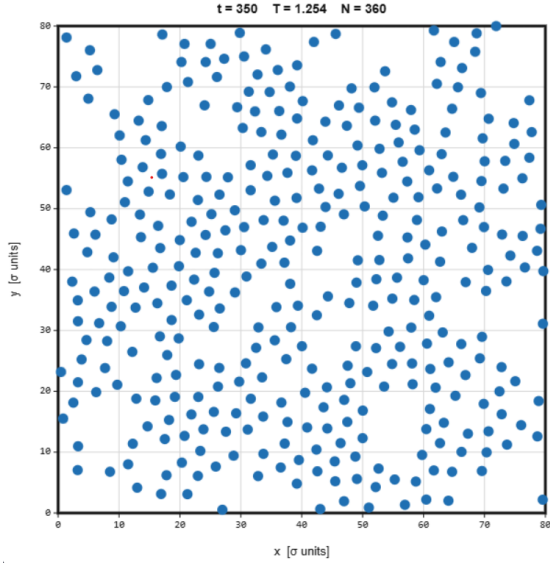
But a wide set of parameters leads to phase transitions from gas-like to fluid-like with local interaction and then further to solid-like long range crystallization. To test the possible parameter space and its crystallization properties, all parameters were set to 1 which leads to crystallization as expected although the phase transition temperatures are slightly different.

If the drag coefficient γ is below 0.3 no crystallization or local long term interaction is observable. For σ , which represents the distance of the potential well away from the center, values below 0.8 lead to spontaneous breakdown of local clusters and ejection of individual edge particles is observable.

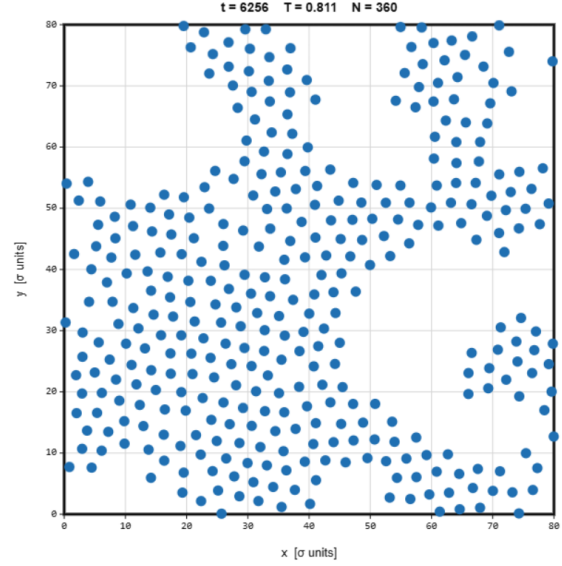
For the potential bond strength ϵ no crystallization or short range order is occurs for a value above 3 if γ and σ are held at 1, but an increase in both leads again to crystalline structures.

Why this system behavior emerges is not fully understood, since a deeper potential well should intuitively make crystallization easier. This may be due to computational limitations that were not further investigated within the scope of this project.

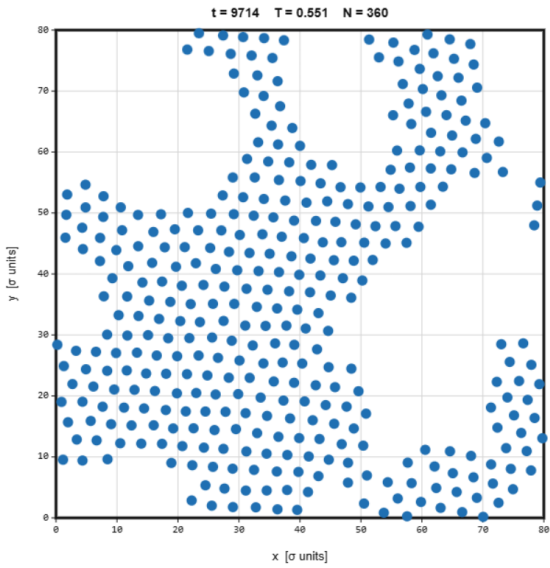
To fully describe and evaluate this system, a parameter space analysis would be necessary but was outside the scope for this project.



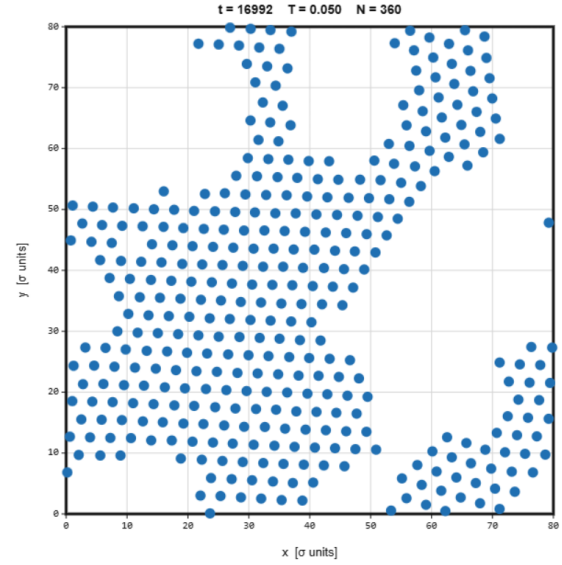
(a) Weakly coupled particles at $t = 350$ with $T = 1.254$



(b) Local short range order at $t = 6256$ with $T = 0.811$



(c) Strong coupling with crystal defects still present at $t = 9714$ and $T = 0.551$



(d) Fully relaxed crystalline structure with long range coupling and crystal defect annealing at $t = 16992$ and $T = T_{min} = 0.05$

Figure 4.1: Snapshots of the particle configuration during the crystallisation process as the system is gradually cooled. The transition from a disordered configuration towards an ordered crystalline structure can be observed.

Chapter 5

Analysis

To evaluate the correctness of the phase transitions and crystallization, visual inspection is not enough. Validation measures are the mean velocity $\langle v \rangle$, the mean of velocity squared $\langle v^2 \rangle$ and the radial distribution function $g(r)$.

5.1 Mean Velocity Validation

In various parameter configurations, $\langle v \rangle$ was measured during the simulation and cooling of the system, and stays stable between 10^{-4} and 10^{-2} . With a mean-zero Gaussian distribution underlying the Brownian motion, no direction of velocity should be preferred, which is reflected in this result. For the simulation shown in figure 4.1, $\langle v \rangle = -1.76 \times 10^{-3}$. This result alone is not sufficient to validate the physical correctness of the simulation, but it does confirm that the system does not exhibit any non-physical preferred direction of motion.

5.2 Mean Squared Velocity Validation

From the equipartition theorem, we know that as the system reaches thermal equilibrium it should approach a value of $\langle v^2 \rangle = 2k_B T/m = 2T$ for our simplified simulation. As seen in figure 5.1, the system starts out highly chaotic and, as it is cooled down, approaches the theoretical value closely. With T_{min} set to 0.05, the theoretical value is $\langle v_{theory}^2 \rangle = 0.10$. The achieved value for the presented simulation lies at $\langle v_{result}^2 \rangle = 0.1083$, a discrepancy of 8.3%. This discrepancy likely stems from the semi-implicit Euler integration scheme, which does not fully resolve the rapidly changing forces present during close-range particle encounters. Nevertheless, since this model is only an approximation intended to capture macroscopic behavior rather than exact microscopic dynamics, the discrepancy is considered to lie within an acceptable physical range.

5.3 Radial Distribution Validation

The final and most decisive validation of the simulation is the radial distribution function. In figure 5.2, peaks appear at the marked theoretical lattice-shell distances, where a perfect crystalline structure would place its particles. Across the three subplots, a clear sharpening of these peaks is visible as they align increasingly closely with the predicted positions. In figure 5.2a, short-range coupling is visible, while long-range order is still close to random, reflected in $g(r)$ remaining close to unity at larger separations. As the system equilibrates, the peaks sharpen and align more closely with theory, while the regions between them become increasingly depopulated. Short-range structure eventually overlaps almost perfectly with theory, while further long-range peaks become clearly pronounced. This is the strongest validation result presented, providing clear structural evidence of a crystalline configuration.

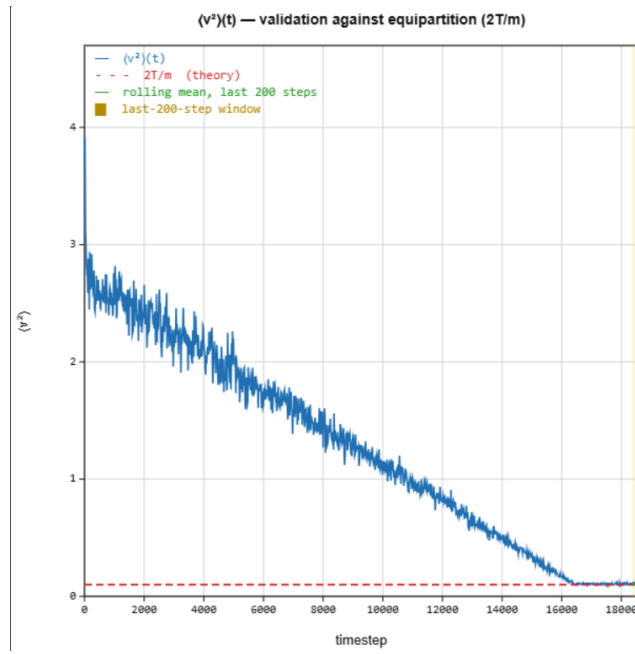
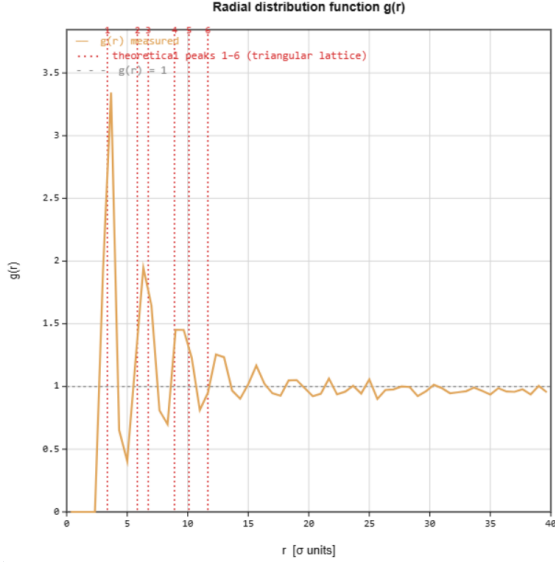
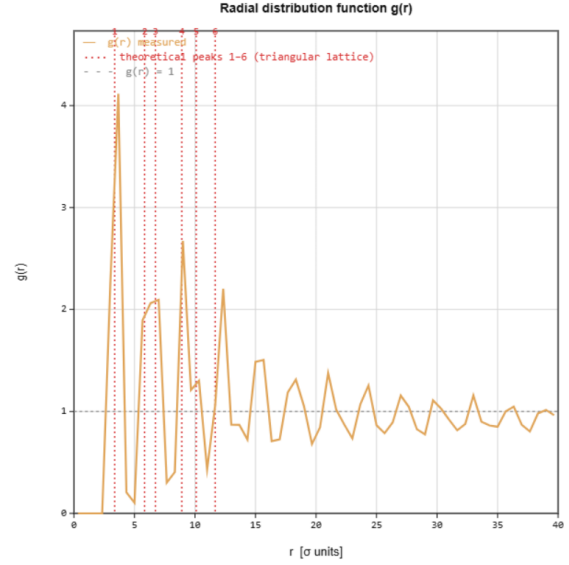


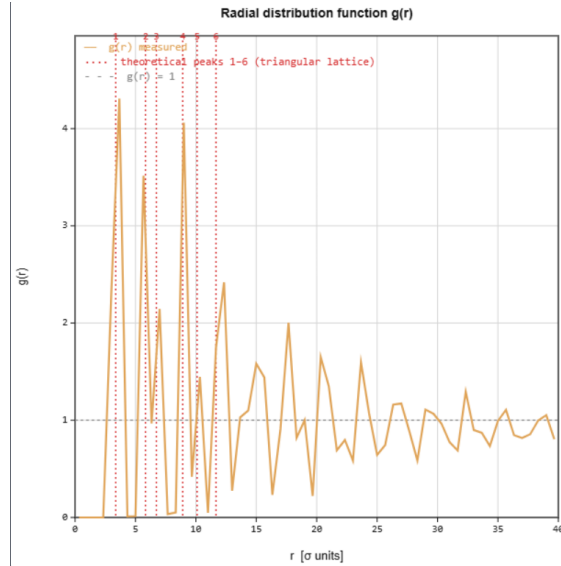
Figure 5.1: Steady decrease of $\langle v^2 \rangle$ over the simulation runtime.



(a) $g(r)$ at $T = 0.811$ showing local coupling and weak long range interaction.



(b) This $g(r)$ plot at $T = 0.551$ shows defined peaks and long range coupling.



(c) The final $g(r)$ at $T = 0.05$ shows very clearly defined peaks and definite long range coupling

Figure 5.2: Three radial distribution plots at different times and temperatures of the simulated system showing short and long range coupling while the number of particles between peaks decreases to almost zero

Chapter 6

Conclusion

This project set out to model the fluid-solid phase transition using a coarse-grained, agent-based simulation combining Lennard-Jones interactions with Langevin dynamics. Through gradual cooling of the system, a clear transition from a disordered, gas-like state to a locally and eventually long-range ordered crystalline structure was observed, including the emergence and subsequent healing of crystal defects during the solidification process.

The physical correctness of this behavior was supported by three independent validation measures. The mean velocity remained close to zero throughout the simulation, consistent with an unbiased stochastic thermostat. The mean squared velocity approached the theoretical equipartition prediction of $2T$ within a discrepancy attributable to the limitations of the integration scheme rather than a fundamental error in the model. Most convincingly, the radial distribution function showed clear, sharpening peaks aligning with the theoretical lattice spacing as the system cooled, providing direct structural evidence of crystallization.

Some aspects of the system's behavior across parameter space, particularly the unexpected loss of crystallization at large bond strengths ϵ , remain incompletely understood and would benefit from a more systematic parameter study. Extending the model to include energy-dependent bonding, three-dimensional interactions, or a more accurate integration scheme would be natural directions for future work. Nevertheless, the results presented here demonstrate that even a simplified, coarse-grained implementation is sufficient to reproduce the qualitative and, to a reasonable degree, quantitative signatures of a genuine fluid-solid phase transition.